SABANCI UNIVERSITY
CS 401 – Computer Architectures
Fall 2025

CLASS SCHEDULE	INSTRUCTOR	TEACHING ASSISTANTS
11:40-12:30, Wed in FASS G018; 10:40-12:30, Thu in FASS G018; 12:40-15:30, Fri in FENS 1033;	Özcan Öztürk Office: FENS 2061, phone: 7038 Office Hr: Thu 13:00-15:00 & Appt. ozcan.ozturk@sabanciuniv.edu	Arda Özgün, arda.ozgun@sabanciuniv.edu Bartu Orhun, bartu.orhun@sabanciuniv.edu Ekin Nalbantoğlu, ekinn@sabanciuniv.edu Emirhan Özdemir, emirhano@sabanciuniv.edu Office Hours: With Appointment

COURSE DESCRIPTION

This is an introductory course on computer architectures which is about the basic concepts and techniques that are fundamental for modern computers such as datapath design, pipelining, memory hierarchy, cache, and virtual memory. Topics include: Instruction set design, computer arithmetic, controller and datapath design, cache and memory systems, input-output systems, interrupts and exceptions, pipelining, performance.

PREREQUISITE

CS 303 Logic and Digital System Design

TEXTBOOK & RECOMMENDED BOOK

- David Money Harris, Sarah L. Harris, *Digital Design and Computer Architecture, 2nd ed.* Morgan Kaufmann, 2013. (Textbook)
- David Patterson, John L. Hennessy, *Computer Organization and Design: The Hardware/Software Interface*, *4th ed*. Morgan Kaufmann, 2012. (Recommended)

GRADING

Quizzes:	10% - 3-7 Quizzes
Labs/Recitations:	20% - 6 Recitation Assignments
Midterm:	30%, Date: TBD
Final:	40%, Date: TBD

Students who fail to meet all of the following requirements will fail the course:

- Score at least 35/100 on the midterm exam
- Score at least 35/100 on the final exam
- Score at least 60% average on lab/recitation assignment grades
- Attend at least 4 lab/recitation assignments (do not miss more than 2 labs/recitations)

ASSIGNMENT GRADING POLICY

No late submission is accepted (no exception). No recitation attendance means recitation assignment grade is zero (including preliminary part). For submitting the recitation work a student must be present in the recitation sessions (students can only attend their section recitation). If a student does not submit a preliminary report s/he only loses preliminary report points.

MAKE-UP POLICY

There will be no make-up for quizzes and labs. Students automatically get 0 (zero) from the respective assignment grade if any of them is missed. Make-up is only allowed for the midterm and final examinations to those with an official health report / official permission on the date of the exam in question. Make-up examinations may be written and/or oral.

ACADEMIC INTEGRITY

If you present someone else's work as yours this would be considered as plagiarism. If a submission is derived from another one by partially changing some parts, this is also plagiarism. When a plagiarism case is detected, we will send the work with the names of the students to the Dean's Office for disciplinary action. You cannot use ChatGPT code (or any other generative artificial intelligence tool), it is classified as plagiarism. Note that there are tools which are capable of detecting ChatGPT code. There will be no tolearance whatsoever. Please see Sabancı University academic integrity page for more details:

https://www.sabanciuniv.edu/en/academic-integrity-statement

CS401 Schedule (tentative)

WEEK	BEGINS	TOPICS COVERED	READINGS	ACTIVITIES
1	29/09/25	Course overview; introduction to computer architecture & assembly language, MIPS assembly language.	Digital Design and Computer Architecture 6.1	
2	06/10/25	MIPS assembly language (cont'd) Oct. 9-10: add/drop period.	DDCA 6.2-6.3	
3	13/10/25	Machine language, programming in MIPS. Sat, Oct. 18: Wed, Oct. 29 Class Makeup (W 11:40-UC G30)	DDCA 6.4-6.5	Assignment 1 **Group 1**
4	20/10/25	MIPS programs, addressing modes	DDCA 6.6-6.7	Assignment 1 **Group 2**
5	27/10/25	Compile, assembling, loading, linking, pseudo-instructions. Exceptions, floating point numbers Wed, Oct. 29 No Class	DDCA 7.1-7.3	Assignment 2 **Group 1**
6	03/11/25	Microarchitecture, single-cycle processor	DDCA 7.6-7.7	Assignment 2 **Group 2**
7	10/11/25	Single-cycle processor (cont'd), performance analysis. Adding new instructions. Nov 10 - Dec 5: withdraw period.	DDCA 7.5	Assignment 3 **Group 2**
8	17/11/25	Pipelined processor.	DDCA 7.8	Assignment 3 **Group 1**
9	24/11/25	Hazards, forwarding. Midterm – TBD		Midterm
10	01/12/25	Forwarding (cont'd), control hazards; Advanced microarchitectures.	DDCA 8.1-8.3	Assignment 4 **Group 1&2**
11	08/12/25	Memory systems: technologies, performance.	DDCA 8.3	
12	15/12/25	Caches	DDCA 8.4	Assignment 5 **Group 1&2**
13	22/12/25	Virtual memory	DDCA 8.5-8.6	Assignment 6 **Group 2**
14	29/12/25	VM-paging and TLB. Intro to I/O. Memory mapped I/O, parallel I/O; Serial I/O w/ SPI, UART; Timers & interrupts, Analog I/O, character LCDs, DC motors, I/O for PCs	DDCA 8.6-8.7	Assignment 6 **Group 1**
15	30/12/25	Classes End. No Class. Finals period 4-13 Jan 2026. Final – TBD		Final